Cascaded optical nonlinearities: Microscopic understanding as a collective effect

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A general microscopic theory that treats cascading of a wide range of nonlinear optical processes as a collective effect is developed. Important practical implications are discussed. © 1997 Optical Society of America [S0740-3224(97)01907-3]

1. INTRODUCTION

Third-order optical nonlinearities, whose potential for optical processing and switching has been understood for many years, have not found wide applications for the simple reason that typical $\chi^{(3)}$ is too small to make the devices based on it practical. Recently, however, significant progress has been made to change this unfavorable situation by use of cascaded second-order nonlinear processes in order to achieve the effective third-order nonlinearity. This method, considered theoretically in Ref. 1 and first realized in Refs. 2 and 3, suggests the use of parametric interaction between fundamental and second-harmonic waves in order to obtain effective $\chi^{(3)}(\omega; -\omega, \omega, \omega)$ proportional to $\chi^{(2)}(\omega; -\omega, 2\omega) \times \chi^{(2)}(2\omega; \omega, \omega)$. In subsequent years, cascaded nonlinearity was successfully implemented in the waveguides,[4,5] all-optical switching with cascading was obtained,[6,7] and all-optical cascaded devices, such as Mach–Zander interferometers were fabricated and tested.[8,9] Proposals have been made for generating optical solitons with cascaded nonlinearity.[10] More recently, a number of cascading schemes that use surface-emitting second-harmonic generation in a resonator have been proposed.[11]

Simultaneously, a completely different route to $\chi^{(3)}$ enhancement was taken by the developers of all-optical processing devices that use the self-electro-optic-effect-device[12] (SEED), where the optical signal is first detected and then modulated, with the possibility of amplification of the low-frequency signal. A closer look at the SEED reveals that its nonlinear action is nothing but cascading, but this time consisting of subsequent application of optical rectification (detection), which can be described by the imaginary part of the effective second-order susceptibility, $\mathrm{Im}[\chi^{(2)}(0; -\omega, \omega)]$, and the linear electro-absorption effect, described by $\mathrm{Im}[\chi^{(2)}(\omega; \omega, 0)]$. Similarly, it is not difficult to see that the photorefractive effect can also be considered as a sequence of detection, $\mathrm{Im}[\chi^{(2)}(0; -\omega, \omega)]$, and linear electro-optic modulation, $\mathrm{Re}[\chi^{(2)}(\omega; \omega, 0)]$. More generally, it was shown in Refs. 13 and 14 that a whole range of charge transport and other Coulomb-assisted nonlinearities in, for example, semiconductor quantized structures, can be considered as cascading.

In this paper we develop a very simple yet general microscopic theory of cascading effects, revealing what is, in fact, the collective nature of the cascading effect. We then evaluate the enhancement produced by the cascading in a number of different third-order nonlinear processes, and we make important practical conclusions.

2. MACROSCOPIC THEORY OF CASCADING

Consider a general third-order nonlinear process [Fig. 1(a)] that involves the interaction of four electro-magnetic waves:

$$\mathbf{E}_n(\mathbf{r}, t) = \mathscr{E}_n \tilde{\mathbf{e}}_n \exp[i(\pm k_n z - \omega_n t)]\psi_n(x, y) + \text{c.c.}; \quad n = 1, 2, 3, 4, \tag{1}$$

where $\psi_n(\mathbf{r})$ are the normalized mode-shape functions (which are used for the case of waveguide propagation, whereas for the case of plane waves they can be assumed to be equal to unity), and $\tilde{\mathbf{e}}_n$ are the unit vectors denoting the polarization direction. For the purpose of simplicity we shall not carry the polarization direction through the rest of the paper; we just assume that the proper components of the nonlinear susceptibility tensors are used. Also, we consider only collinearly propagating waves; the basic conclusions of this paper do not change in the more general case. The plus and minus signs in front of the wave vector in Eq. (1) indicate that the possibility of a counterpropagation is considered.

The nonlinear interaction leads to the appearance of the third-order polarization:

$$\begin{aligned} P_4^{(3)}(\omega_4 &= \omega_1 + \omega_2 + \omega_3) \\ &= \epsilon_0 \chi^{(3)}(\omega_1 + \omega_2 + \omega_3; \omega_1, \omega_2, \omega_3)\mathscr{E}_1\mathscr{E}_2\mathscr{E}_3 \\ &\quad \times \exp(-i\omega_4 t)\psi_1(x, y)\psi_2(x, y)\psi_3(x, y) \\ &\quad \times \exp[i(k_1 + k_2 + k_3)z], \end{aligned} \tag{2}$$

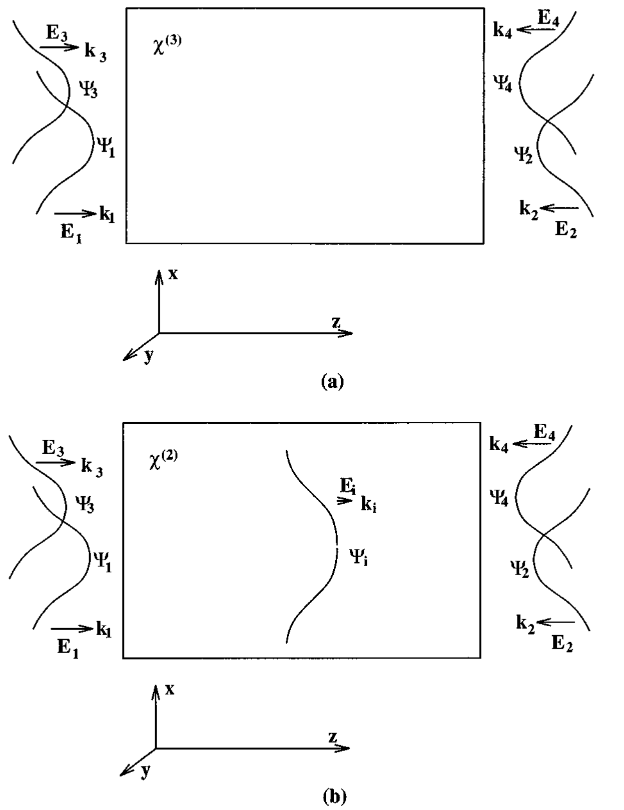

Fig. 1. General four-wave mixing in (a) $\chi^{(3)}$ medium, (b) $\chi^{(2)}$ medium.

where the third-order susceptibility[15] is

$$\chi^{(3)}(\omega_1 + \omega_2 + \omega_3;\, \omega_1,\, \omega_2,\, \omega_3) = \sum_{jkl} \frac{Ne^4 \mathbf{r}_{ij}\mathbf{r}_{jk}\mathbf{r}_{kl}\mathbf{r}_{li}}{\epsilon_0(E_{ji} - \hbar\omega_1)[E_{ki} - \hbar(\omega_1 + \omega_2)][E_{li} - \hbar(\omega_1 + \omega_2 + \omega_3)]}, \tag{3}$$

and the matrix elements are

$$\mathbf{r}_{ij} = \langle i|\mathbf{r}|j\rangle. \tag{4}$$

Note that the presence of complex-conjugate components in Eq. (1), as well as the inclusion of counterpropagating waves, means that all the frequencies and wave vectors in Eq. (2) and after can be both positive and negative, i.e., $\omega_n = \pm|\omega_n|$, $k_n = \pm|\mathbf{k}_n|$. For example, if $\omega_1 = \omega_2 = \omega_3 = |\omega|$, Eq. (2) describes third-harmonic generation, whereas if $\omega_1 = \omega_3 = |\omega|$; $\omega_3 = -|\omega|$, then Eq. (2) describes Kerr-type nonlinearity.

The propagation equation for the nonlinear wave is

$$\nabla^2\mathbf{E}_4 - \frac{n^2}{c^2}\frac{\partial^2\mathbf{E}_4}{\partial t^2} = \frac{1}{\epsilon_0 c^2}\frac{\partial^2 P_4^{(3)}}{\partial t^2}, \tag{5}$$

which in the slow-varying amplitude approximation becomes, upon substitution of Eq. (2) and integration over the transverse coordinates x and y,

$$\frac{\mathrm{d}\mathscr{E}_4}{\mathrm{d}z} = i\,\frac{\omega_4}{2cn}\,\kappa_{1234}\chi^{(3)}(\omega_4;\, \omega_1,\, \omega_2,\, \omega_3)\mathscr{E}_1\mathscr{E}_2\mathscr{E}_3 \times \exp[(k_1 + k_2 + k_3 - k_4)z], \tag{6}$$

where the overlap integral is

$$\kappa_{1234} = \frac{\int \psi_1\psi_2\psi_3\psi_4 \mathrm{d}x\mathrm{d}y}{\int {\psi_4}^2 \mathrm{d}x\mathrm{d}y}. \tag{7}$$

To implement the third-order process using only second-order susceptibility [Fig. 1(b)], we first need to create a nonlinear polarization at the intermediate frequency $\omega_i = \omega_1 + \omega_2$:

$$P_i^{(2)}(\omega_i = \omega_1 + \omega_2) = \epsilon_0\chi^{(2)}(\omega_1 + \omega_2;\, \omega_1,\, \omega_2)\mathscr{E}_1\mathscr{E}_2 \times \exp(-i\omega_i t)\psi_1(x,\, y)\psi_2(x,\, y) \times \exp[i(k_1 + k_2)z], \tag{8}$$

where the second-order susceptibility is

$$\chi^{(2)}(\omega_1 + \omega_2;\, \omega_1,\, \omega_2) = \sum_{jk} \frac{Ne^3\mathbf{r}_{ij}\mathbf{r}_{jk}\mathbf{r}_{ki}}{\epsilon_0(E_{ji} - \hbar\omega_1)[E_{ki} - \hbar(\omega_1 + \omega_2)]}. \tag{9}$$

The nonlinear polarization $P^{(2)}(\omega_i)$ gives rise to the electromagnetic wave at the intermediate frequency $E_i(\omega_i)$, whose amplitude can be determined from the wave equation

$$\nabla^2 E_i - \frac{n^2}{c^2}\frac{\partial^2 E_i}{\partial t^2} - \frac{n^2}{c^2\tau_c}\frac{\partial E_i}{\partial t} = \frac{1}{\epsilon_0 c^2}\frac{\partial^2 P_i}{\partial t^2}, \tag{10}$$

where the third term on the left-hand side is included only in the case in which the intermediate wave $\mathbf{E}_i$ is confined in a cavity with lifetime τ_c.

From Eq. (10), depending on the relation between the physical size of the system L and the wavelength of the intermediate wave in the medium $\lambda_i = 2\pi/nk_i$, it is not difficult to obtain the slowly varying amplitude of the intermediate wave in the following two regimes.

- Long-wavelength approximation ($\omega_1 \approx -\omega_2$), or $L \ll \lambda_i$. The response in this case [Fig. 2(a)] is entirely local, and the transverse shape of the low-frequency mode simply follows the shape of the nonlinear polarization:

$$\psi_i(x,\, y) = \psi_1(x,\, y)\psi_2(x,\, y), \tag{11}$$

$$\mathscr{E}_i = -\frac{1}{\epsilon_r(\omega_i)}\,\chi^{(2)}(\omega_1 + \omega_2;\, \omega_1,\, \omega_2) \times \mathscr{E}_1\mathscr{E}_2 \exp[i(k_1 + k_2)z]. \tag{12}$$

- $L \gg \lambda$. In this case [Fig. 2(b)] the slowly varying amplitude approach shall be used to obtain the expression for the intermediate wave, assuming that the two incoming waves, E_1 and E_2, are not depleted. Then

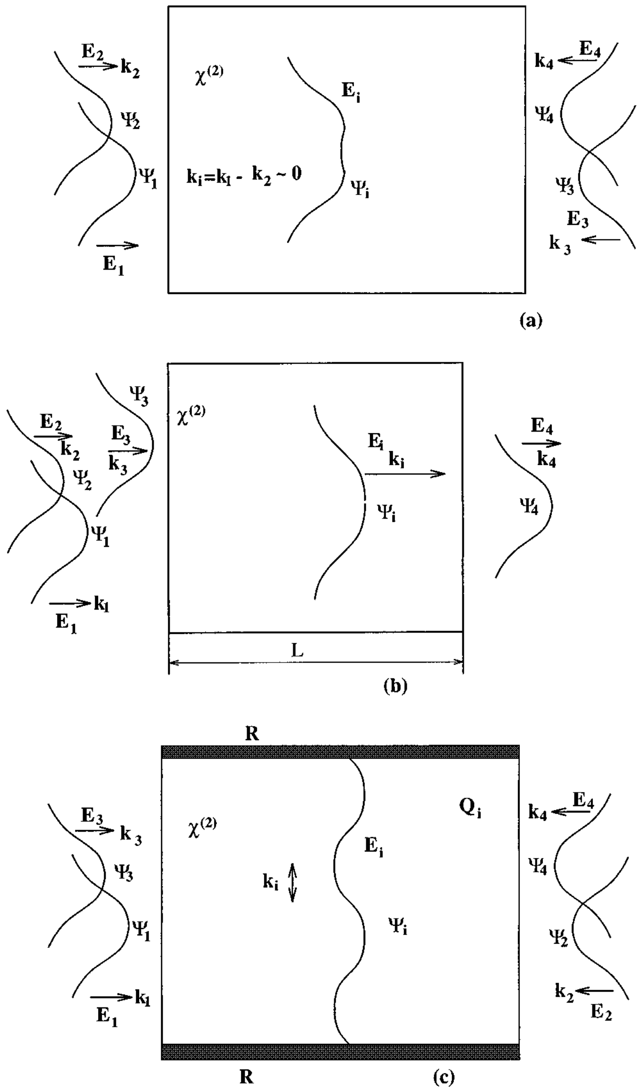

Fig. 2. Geometry of cascading: (a) Long–intermediate-wavelength limit, (b) collinear, and (c) transverse with the cavity for the intermediate wave.

$$\mathscr{E}_i(z) = \frac{\omega_i}{2cn}\chi^{(2)}(\omega_1 + \omega_2;\ \omega_1,\ \omega_2) \times \mathscr{E}_1\mathscr{E}_2\kappa_{12i}\frac{1 - \exp(-i\Delta kz)}{\Delta k} \times \exp[i(k_1 + k_2 - k_i)z], \tag{13}$$

where

$$\Delta k = k_1 + k_2 - k_i, \tag{14}$$

$$\kappa_{12i} = \frac{\int \psi_1\psi_2\psi_i \mathrm{d}x\mathrm{d}y}{\int \psi_i^2 \mathrm{d}x\mathrm{d}y}. \tag{15}$$

Consider the phase-mismatch term $[1-\exp(-i\Delta kz)]/(\Delta k)$, which is, of course, distance dependent. Nevertheless, for the simple purpose of comparing nonlinearities, it can be replaced by its distance-averaged value, which, in turn, depends on the relation between the length L and the coherence length $l_{\mathrm{coh}} = 2\pi/\Delta k$:

1. $L \ll l_{\mathrm{coh}}$;

$$\left\langle \frac{\omega_i}{2cn}\frac{1 - \exp(-i\Delta kz)}{\Delta k}\right\rangle \approx i\frac{\omega_i}{2cn}\frac{L}{2} = i\frac{\pi L}{2\lambda_i\epsilon_r(\omega_i)} = i\frac{\omega_i\tau_p}{2\epsilon_r(\omega_i)}, \tag{16}$$

where $\tau_p = Ln/2c$ is the average propagation time in the material.

2. $L \gg l_{\mathrm{coh}}$; in this case,

$$\left\langle \frac{\omega_i}{2cn}\frac{1 - \exp(-i\Delta kz)}{\Delta k}\right\rangle \approx \frac{\omega_i}{2cn}\frac{1}{\Delta k} = \frac{l_{\mathrm{coh}}}{2\lambda_i\epsilon_r(\omega_i)} = \frac{\omega_i\tau_{\mathrm{coh}}}{2\pi\epsilon_r(\omega_i)}, \tag{17}$$

where $\tau_{\mathrm{coh}} = l_{\mathrm{coh}}n/2c$ is the coherence time for the nonlinear process.

- Assuming that there is a resonant cavity [Fig. 2(c)] for the intermediate field,[11] it is easy to obtain

$$\mathscr{E}_i(z) = i\frac{\omega_i\tau_c}{n^2}\chi^{(2)}(\omega_1 + \omega_2;\ \omega_1,\ \omega_2)\mathscr{E}_1\mathscr{E}_2\kappa_{12i} \times \exp[i(k_1 + k_2)z]. \tag{18}$$

Before proceeding further, let us attempt to unite various geometries. Comparing Eqs. (16)–(18), one can easily see that they all include some effective lifetime of the electromagnetic wave at the intermediate frequency. In the case of the standing wave the lifetime is determined by all cavity losses, while in the case of the propagating wave it is limited either by the photon exiting the material or by getting out of phase with polarization, whichever happens first. One can now introduce a quality factor of the experiment Q_i, where for the long-wavelength case, $Q_i = 1$, for the resonant cavity, $Q_i = \omega_i\tau_c$, and for the collinear propagation case, Q_i is the smaller one of $\omega_i\tau_p$ and $\omega_i\tau_{\mathrm{coh}}$. Also, another new parameter, β, can be introduced as

$$\mathscr{E}_i(z) = \chi^{(2)}(\omega_1 + \omega_2;\ \omega_1,\ \omega_2) \times \beta\kappa_{i34}^{-1}\mathscr{E}_1\mathscr{E}_2\exp[i(k_1 + k_2 - k_i)z], \tag{19}$$

where

$$\kappa_{i34} = \frac{\int \psi_i\psi_3\psi_4 \mathrm{d}x\mathrm{d}y}{\int \psi_i^4 \mathrm{d}x\mathrm{d}y}. \tag{20}$$

We finally obtain

$$\beta = \frac{M}{\epsilon_r(\omega_i)}Q_i\kappa_{12i}\kappa_{i34}, \tag{21}$$

where the factor M is either real or imaginary with an absolute value of the order of unity.

Now, the interaction of the intermediate field with the remaining wave $\mathbf{E}_3$ causes the nonlinear polarization of frequency $\omega_4 = \omega_i + \omega_3$:

$$P_4^{(2+2)}(\omega_4 = \omega_1 + \omega_2 + \omega_3) = \epsilon_0 \chi^{(2)}(\omega_i + \omega_3;\, \omega_i,\, \omega_3)\mathscr{E}_i \mathscr{E}_3 \times \exp(-i\omega_4 t)\psi_i(x, y)\psi_3(x, y) \times \exp[i(k_i + k_3)z] = \beta\epsilon_0 \chi^{(2)}(\omega_i + \omega_3;\, \omega_i,\, \omega_3) \times \chi^{(2)}(\omega_1 + \omega_2;\, \omega_1,\, \omega_2) \times \kappa_{i34}^{-1}\mathscr{E}_1\mathscr{E}_2\mathscr{E}_3\psi_i(x, y)\psi_3(x, y) \times \exp[i(k_1 + k_2 + k_3)z], \quad (22)$$

where

$$\chi^{(2)}(\omega_i + \omega_3;\, \omega_i,\, \omega_3) = \sum_{kl} \frac{Ne^3 \mathbf{r}_{ij}\mathbf{r}_{jk}\mathbf{r}_{ki}}{\epsilon_0[E_{li} - \hbar(\omega_1 + \omega_2 + \omega_3)][E_{ki} - \hbar(\omega_1 + \omega_2)]}. \quad (23)$$

Substituting Eq. (22) into Eq. (5), one obtains, after integration over the transverse coordinates x and y,

$$\frac{\mathrm{d}\mathscr{E}_4}{\mathrm{d}z} = i\,\frac{\omega_4}{2cn}\,\beta\chi^{(2)}(\omega_i + \omega_3,\, \omega_i,\, \omega_3) \times \chi^{(2)}(\omega_1 + \omega_2;\, \omega_1,\, \omega_2)\mathscr{E}_1\mathscr{E}_2\mathscr{E}_3 \times \exp[(k_1 + k_2 + k_3 - k_4)z]. \quad (24)$$

Therefore, after comparing Eq. (24) with Eq. (6), we can introduce the expression for the artificial third-order susceptibility $\chi^{(2+2)}$:

$$\chi^{(2+2)}(\omega_1 + \omega_2 + \omega_3;\, \omega_1,\, \omega_2,\, \omega_3) = \chi^{(2)}(\omega_1 + \omega_2;\, \omega_1,\, \omega_2) \times \chi^{(2)}(\omega_i + \omega_3;\, \omega_i,\, \omega_3)\,\frac{M}{\epsilon_r(\omega_i)}\,Q_i\,\frac{\kappa_{12i}\kappa_{i34}}{\kappa_{1234}}. \quad (25)$$

Note that $\chi^{(2)}$ enter Eq. (25) as a product and not as a square of magnitude; therefore, by changing the phase of $\chi^{(2)}$ by $\pi/4$ (which can be done in the vicinity of a resonance) we can change character of effective third-order susceptibility from absorptive to refractive and vice versa, an important fact to consider. It is instructive to represent cascading of the enhancement as a product of two components,

$$\eta_{\text{casc}} = \frac{\chi^{(2+2)}}{\chi^{(3)}} = \eta_{\text{mat}}\eta_{\text{geo}}, \quad (26)$$

where the material (intrinsic) component is

$$\eta_{\text{mat}} = \frac{\chi^{(2)}(\omega_i;\, \omega_1,\, \omega_2)\chi^{(2)}(\omega_4;\, \omega_i,\, \omega_3)}{\epsilon_r(\omega_i)\chi^{(3)}(\omega_4;\, \omega_1,\, \omega_2,\, \omega_3)} \quad (27)$$

and the geometrical (field) component is

$$\eta_{\text{geo}} = MQ_i\,\frac{\kappa_{12i}\kappa_{i34}}{\kappa_{1234}}, \quad (28)$$

which is equal to the Q_i factor for all practical purposes.

To evaluate the intrinsic component within an order of magnitude for the general system, one performs averaging over all the transitions involved and introduces an average dipole $e\bar{d}$, $\bar{d}$ being within an order of magnitude of the wave-function extent, and an average detuning at the intermediate frequency $\langle E_{ki} - \hbar\omega_i\rangle = \overline{\delta E}$. Then, it is not difficult to obtain

$$\eta_{\text{mat}} = \frac{Ne^2\bar{d}^2}{\epsilon_0\epsilon_r(\omega_i)\overline{\delta E}}. \quad (29)$$

If one is dealing with a resonant process, effective detuning of $\overline{\delta E}$ in Eq. (29) is changed into broadening $i\Gamma$, and, if one is dealing with a solid-state material excited above the fundamental gap, then, instead of $N/\overline{\delta E}$, one uses an effective density of states $g(E)$. It is not difficult to observe that the intrinsic factor is

$$\eta_{\text{mat}} = \frac{Ne^2\bar{d}^2}{\epsilon_0\overline{\delta E}\,\epsilon_r(\omega_i)} \approx \frac{\chi^{(1)}(\omega_i)}{\epsilon_r(\omega_i)} \approx \frac{\epsilon_r(\omega_i) - 1}{\epsilon_r(\omega_i)}; \quad (30)$$

i.e., this intrinsic factor is typically somewhat below unity, and the enhancement is entirely due to the Q_i factor. Using resonance at the intermediate frequency does not improve the cascading enhancement, since an increase of $\chi^{(2)}$ in the numerator of Eq. (27) is compensated by the increase of $\epsilon_r(\omega_i)$ in the denominator.

3. MICROSCOPIC ORIGIN OF CASCADING ENHANCEMENT

So, what is a physical, or, more precisely, a microscopic origin of the enhancement? Let us go back to the expression for the third-order susceptibility [Eq. (3)] and consider the Feynman diagram in Fig. 3. It is easy to follow from density-matrix equations that interaction of the first wave, $\mathbf{E}_1$; causes the appearance of the off-diagonal matrix element $\rho_{ji}\exp(-i\omega_1 t)$ that has a lifetime determined from the uncertainty principle $\tau_{ji} = (E_{ji}/\hbar - \omega_1)^{-1}$. Following that, interaction with the second wave, $\mathbf{E}_2$, leads to the appearance of the off-diagonal element $\rho_{ki}\exp[-i(\omega_1 + \omega_2)t]$ whose lifetime is $\tau_{ki} = (E_{ki}/\hbar - \omega_1 - \omega_2)^{-1}$. Finally, the third wave, $\mathbf{E}_3$, gives rise to the off-diagonal element $\rho_{li}\exp(-i\omega_4 t)$ whose lifetime is $\tau_{li} = (E_{li}/\hbar - \omega_4)^{-1}$. Notice that matrix elements $\rho_{ji}\exp(-i\omega_1 t)$ and $\rho_{li}\exp(-i\omega_4 t)$ are resonantly coupled with the fields $\mathbf{E}_1$ and $\mathbf{E}_4$, respectively, while matrix element $\rho_{ki}\exp[-i(\omega_1 + \omega_2)t]$ is not.

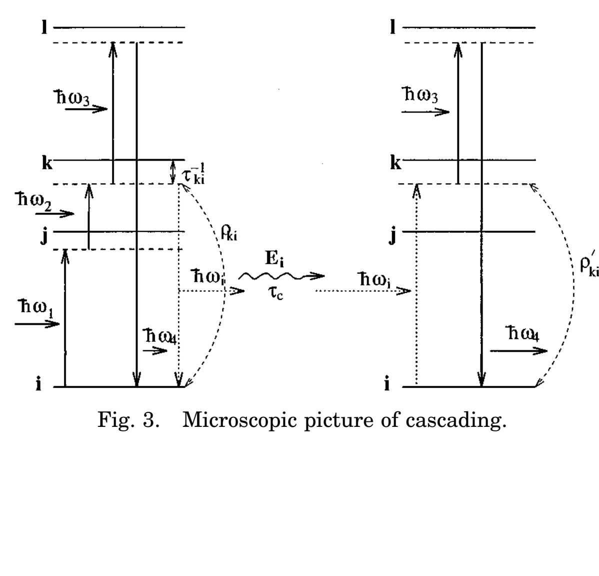

Fig. 3. Microscopic picture of cascading.

Let us see what happens to the lifetime of the matrix element $\rho_{ki} \exp[-i(\omega_1 + \omega_2)t] = \rho_{ki} \exp(-i\omega_i t)$ in the cascading scheme. First, the dipole moment at the intermediate frequency appears:

$$\mathbf{p}_i(\omega_i) = e\mathbf{r}_{ki}\rho_{ki} \exp(-i\omega_i t). \tag{31}$$

The dipole moments of individual atoms or molecules add up and give rise to the average electric field at intermediate frequency

$$\begin{aligned}\mathbf{E}_i(\omega_i) &= Q_i N \mathbf{p}_i(\omega_i)/\epsilon_0\epsilon_r(\omega_i) \\ &= Q_i N e \mathbf{r}_{ki}\rho_{ki} \exp(-i\omega_i t)/\epsilon_0\epsilon_r(\omega_i).\end{aligned} \tag{32}$$

Next, the intermediate field, produced by all the atoms (hence the collective nature of the cascading effect), acts back on each individual atom and gives rise to the matrix element

$$\begin{aligned}\rho_{ki} \exp(-i\omega_i t) &= e\mathbf{r}_{ik}\mathbf{E}_i(\omega_i)/(E_{ki} - \hbar\omega_i) \\ &= Q_i \frac{Ne^2\mathbf{r}_{ki}\mathbf{r}_{ik}}{\epsilon_0\epsilon_r(\omega_i)(E_{ki} - \hbar\omega_i)} \rho_{ki} \exp(-i\omega_i t) \\ &\sim Q_i \frac{\epsilon_r(\omega_i) - 1}{\epsilon_r(\omega_i)} \rho_{ki} \exp(-i\omega_i t).\end{aligned} \tag{33}$$

So, the magnitude of the off-diagonal matrix element $\rho_{ki} \exp(-i\omega_i t)$ is enhanced by exactly the factor shown in Eqs. (26) and (30). Note that the back action of induced polarization onto the electric field had already been taken into account by including $\epsilon_r(\omega_i)$ into the denominator of Eq. (32). Therefore we can make the following microscopic interpretation of the cascading effect: by resonant coupling of the material oscillations $\rho_{ki} \exp(-i\omega_i t)$ at the intermediate frequency with the electromagnetic field $\mathbf{E}_i(\omega_i)$, the effective lifetime of these oscillations is enhanced by a factor Q_i.

In other words we can consider a matrix element $\rho_{ki} \exp(-i\omega_i t)$ that, in the absence of coupling to the electromagnetic wave, decays in a very short time $(E_{ki}/\hbar - \omega_i)$. If, on the other hand, the matrix element is coupled to the electromagnetic wave, a photon of frequency ω_i can be emitted and then reabsorbed by one of the atoms, recreating $\rho_{ki} \exp(-i\omega_i t)$. This continuous exchange of energy between the media and the electromagnetic wave prolongs the effective life of the matrix element $\rho_{ki} \exp(-i\omega_i t)$ because the Q of the electro-magnetic wave, with or without cavity, is typically much higher than the Q of atomic transitions. Indeed, even in the case of resonance, the photon lifetime is typically longer than the coherence time of the transition.

Thus, when a coherent excitation at frequency ω_i in the form of the atomic excitation is created by nonlinear interaction of two waves, it is going to decay very fast. However, after being transformed into the electromagnetic field, this excitation can be stored in the form of a field for a much longer period of time and later transformed back into the atomic excitation.

It also makes sense to make a simple analogy with electric circuits. When a circuit has low Q, coupling it to the second circuit of much higher Q raises the overall Q of the system.

4. SPACE-CHARGE NONLINEARITY AS CASCADING

Before attempting to write any conclusions, it is instructive to enlarge the scope of this general treatment of cascading by considering a host of similarly unrelated phenomena: photorefractive effects,[16,17] hetero-nipi superlattices,[18] and a number of other charge-transport nonlinear optical effects.

When two waves with closely spaced frequencies ω_1 and ω_2 are present (Fig. 4), there is a beat-frequency irradiance

$$I(\omega_i = \omega_1 - \omega_2) = \frac{n}{2\eta_0} \mathscr{E}_1\mathscr{E}_2\psi_1\psi_2 \times \exp[i(k_1 - k_2)z - \omega_i t], \tag{34}$$

where $\eta_0 = 377\Omega$ is a free-space impedance. Absorption of the beat-frequency irradiation and subsequent drift and diffusion will create a time- and coordinate-dependent space-charge density. In order to make our order-of-magnitude estimate independent of the nature of the effects leading to the creation of the space-charge field (diffusion with subsequent trapping in case of photorefractive effects, drift in the case of a nipi superlattice, etc.), we just assume the existence of some mean space charge $\rho = \rho(\omega_i)\exp(-i\omega_i t)$ distributed with the mean charge-separation distance d_{sc}. It is easy to see that, in the case of photo-refractive two- and four-wave-mixing experiments, d_{sc} is equal to the half-period of photoinduced grating $\pi|k_1 - k_2|^{-1}$; for the nipi superlattice it is equal to its half-period; for the conjugated polymers, d_{sc} is roughly equal to the length of molecule, and so on. We can now write a rate equation

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = e\alpha(\omega_1)I(\omega_i)/\hbar\omega_i - \frac{\rho}{\tau_r}, \tag{35}$$

where τ_r is the recombination time, and the absorption coefficient is

$$\alpha(\omega_1) \approx \alpha(\omega_2) = \frac{\omega_1}{nc} \sum_j \frac{Ne^2\mathbf{r}_{ij}\mathbf{r}_{ji}}{\epsilon_0\Gamma}. \tag{36}$$

If one deals with absorption into the energy band rather than into the set of discrete states, reduced density of states $g(\hbar\omega_1)$ is used in Eq. (36) in place of N/Γ. Solving Eq. (35) with Eqs. (34) and (36) yields

$$\rho(\omega_i) = \frac{Ne^3\bar{d}^2\mathscr{E}_1\mathscr{E}_2}{2\Gamma\hbar(\tau_r^{-1} - j\omega_i)}. \tag{37}$$

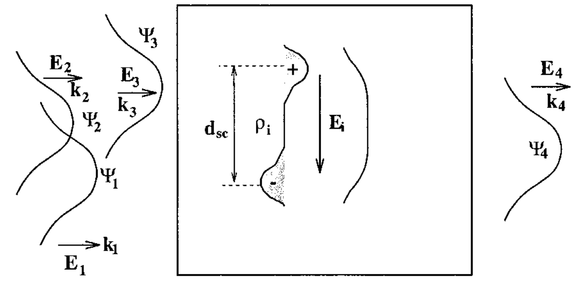

Fig. 4. Space-charge nonlinearity as a special case of cascading.

Assuming that the intermediate frequency is well within the bandwidth of the material ($\omega_i \ll \tau_r^{-1}$), one obtains the space-charge field at the intermediate frequency, which becomes

$$\mathscr{E}_i \approx \frac{\rho(\omega_i)d_{\text{sc}}}{\epsilon_r(\omega_i)\epsilon_0} = \frac{Ne^3\mathbf{r}_{if}\mathbf{r}_{fi}d_{\text{sc}}}{2\epsilon_r(\omega_i)\epsilon_0\Gamma\hbar\tau_r^{-1}}\mathscr{E}_1\mathscr{E}_2, \qquad (38)$$

where we drop the summation since, realistically, absorption takes place only in one transition from initial state i to final state f. Comparing this expression for the intermediate frequency electric field [Eq. (12)], one can formally introduce an effective second-order susceptibility for the space-charge effects,

$$\chi_{\text{sc}}^{(2)} = \frac{Ne^3\mathbf{r}_{if}\mathbf{r}_{fi}d_{\text{sc}}}{2\epsilon_0\Gamma\hbar\tau_r^{-1}}, \qquad (39)$$

and obtain an equation similar to Eq. (12),

$$\mathscr{E}_i = \frac{1}{\epsilon_r(\omega_i)}\chi_{\text{sc}}^{(2)}\mathscr{E}_1\mathscr{E}_2\exp[i(k_1 - k_2)z]. \qquad (40)$$

Comparing Eq. (39) with Eq. (9), one can easily see a substantial enhancement in the strength of intermediate field, which is two-fold:

$$\eta_{\text{sc}} = \frac{\chi_{\text{sc}}^{(2)}}{\chi^{(2)}} \sim \frac{\tau_r}{(\overline{\delta\omega})^{-1}}\frac{d_{\text{sc}}}{\bar{d}}, \qquad (41)$$

where $\overline{\delta\omega} = \overline{\delta\mathbf{E}}/\hbar$ is some average frequency detuning, $\bar{d}$ as mentioned above is of the order of the spatial extent of the wave function, and the completeness of the set of wave functions is used.

First, there is an enhancement associated with the charge accumulation in time. Second, there is an enhancement associated with the generation of giant dipole ed_{sc}. It is also easy to see that, as a result of enhancement of magnitude, the response time of nonlinearity increases, since drift and diffusion take a long time and result in charge separation and trapping, greatly increasing the recombination time.

5. CONCLUSIONS

First, it is worthwhile to note that our derivation, involving interaction of four arbitrary waves, shows that virtually any third-order nonlinear process can be successfully emulated by a cascaded combination of two second-order processes. These processes include self-phase modulation, temporal and spatial soliton propagation, and phase conjugation.

It is interesting to mention that, while cascading has become a hot topic in the past few years, mostly owing to the successful emulation of the Kerr nonlinearity by obtaining an effective $\chi^{(3)}(\omega;\, -\omega,\, \omega,\, \omega)$ by cascading of $\chi^{(2)}(2\omega;\, \omega,\, \omega)$ and $\chi^{(2)}(\omega;\, -\omega, 2\omega)$, cascading of $\chi^{(2)}$ has been known and used successfully for years.

One better-known example is generation of the third harmonic $\chi^{(3)}(3\omega;\, -\omega,\, \omega,\, \omega)$ with cascading of $\chi^{(2)}(2\omega;\, \omega,\, \omega)$ and $\chi^{(2)}(3\omega;\, \omega, 2\omega)$.[19] This method has always been used to obtain UV radiation at 355 nm from the fundamental radiation of the Nd–YAG lasers.

Another well-known example is a SEED in which the optical field is first detected and then the voltage drop across the device changes; this is a rectification process. Subsequently, the photoinduced voltage changes the absorption and the refraction of the material.

A third example is an optical parametric oscillator or amplifier that performs, with $\chi^{(2)}$, the same function that the Raman oscillator performs with $\chi^{(3)}$, but with a much higher efficiency. It is easy to see that the optical parametric process is nothing but cascading with ω_i being the idler frequency. Indeed, at first the idler frequency is produced with $\chi^{(2)}(-\omega_i;\, \omega_2,\, -\omega_1)$, and then it gives rise to the signal with $\chi^{(2)}(\omega_2;\, -\omega_i,\, \omega_1)$, resulting in what is essentially a Raman susceptibility $\chi^{(2+2)}(\omega_2;\, -\omega_1,\, \omega_2,\, \omega_1)$.

The microscopic picture of cascading developed in this paper establishes that the advantage of cascading lies in the collective, or coherent, effect of extending the lifetime of the virtual state by means of coupling that state to an electromagnetic wave. The enhancement produced by cascading is equal to the Q factor of the intermediate electromagnetic wave, which can be introduced independent of whether there is a resonant cavity.

Clearly, there are two distinct ways of cascading, depending on whether the intermediate frequency ω_i is a sum ($\omega_1 + \omega_2$) or a difference ($\omega_1 - \omega_2$) of two signal waves. The sum method (including a degenerate case of second-harmonic generation), obviously leads to much higher Q, but it can place strict demands on phase matching. The difference method leads to lower Q (order of 1), yet large enhancement can be obtained with space-charged nonlinearities, including the photorefractive effect. The drawback is, of course, the slow speed of all such schemes.

Overall, the enhancement provided by cascading is substantial, especially with the resonator, but the potential advantages have to be balanced against difficulties in phase matching. A further enhancement of the cascaded nonlinearity can be achieved by employing various hybrid optoelectronic schemes, where the intermediate field is amplified. In the low-ω_i schemes the intermediate field can be amplified by electronic means; in fact, this is exactly what takes place in the field-effect-transitor SEED.[12] This method can be extended all the way to microwaves. In the high-ω_i schemes the intermediate field can be amplified in a laser amplifier.

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